INCH-POUND

MIL-PRF-23098B(SH)
7 May 2025
SUPERSEDING
MIL-R-23098A(SHIPS)
30 November 1970

PERFORMANCE SPECIFICATION

REGULATOR, LINE VOLTAGE, SINGLE-PHASE AND THREE-PHASE, 400 HERTZ ALTERNATING CURRENT (NAVAL SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general requirements for single-phase and three-phase, 400 Hertz (Hz) alternating current (AC) line voltage regulators intended for naval shipboard application.

1.2 Power rating. The continuous load capacity of the three-phase bank is limited by the maximum line current rating of the single-phase equipment. The equipment is rated at 440 volts (V) root mean square (RMS) and the following power ratings:

a. single-phase 15 kilovolt-amperes (kVA)

b. single-phase 30 kVA

c. three-phase 22.5 kVA

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-27	-	Transformers and Inductors (Audio, Power, and High-Power Pulse), General Specification for
MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-DTL-2036	-	Enclosures for Electric and Electronic Equipment, Naval Shipboard
MIL-PRF-19500	-	Semiconductor Devices, General Specification for

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6110

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current
MIL-STD-1474	-	Noise Limits
MIL-STD-2003	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-338	-	Electronic Reliability Design Handbook
MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume I
MIL-HDBK-472	-	Maintainability Prediction
MIL-HDBK-2084	-	Handbook for Maintainability of Avionic and Electronic Systems and Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 General requirements. The equipment shall be in accordance with the safety requirements of MIL-DTL-917. If any requirement specified herein conflicts with the safety requirements of MIL-DTL-917, the requirements of this specification shall govern.

3.2.1 Vibration. The equipment shall withstand the type I vibration test in accordance with MIL-STD-167-1.

3.2.2 Shock. The equipment shall withstand the high-impact shock tests for grade A, class I, type A, lightweight equipment in accordance with MIL-DTL-901.

3.2.3 Airborne noise. For all conditions of operation specified herein, the equipment shall conform to the acceptance limits for limit A equipment in accordance with MIL-STD-1474.

3.2.4 Electromagnetic interference. The equipment shall be designed to meet the requirements for emission and susceptibility in accordance with MIL-STD-461 requirements: RE101, RE102, RS101, RS103, CS101, CE102, CS114, and CS116.

3.2.5 System protection. The equipment shall be designed to operate within the constraints of the type III electric power system in accordance with MIL-STD-1399-300-1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

3.2.6 Mounting. The equipment shall be designed for mounting in accordance with the equipment mounting requirements of MIL-DTL-917. No more than four bolts shall be required for mounting. Each equipment shall be equipped with cable entrance provisions that shall permit the passage of external cabling through the side panels to the input-output terminal board positions in the wiring compartment. All mounting provisions shall be arranged for easy access and for the adjustment of mounting screws or bolts without the use of special tools and without disassembly of the equipment. Special tools are defined as those tools not listed in the GSA Global Supply Catalog.

3.2.7 Enclosure. Each equipment shall be provided with at least a drip proof protected enclosure constructed in accordance with the enclosure design requirements of MIL-STD-108. If the electrical load powered by the equipment requires a different degree of enclosure protection, such as spray-tight or splashproof, this shall be applied to the equipment as well in accordance with MIL-STD-108.

3.2.8 Insulation. The insulating material temperature rating shall be selected based on the insulating material for bus supports and spacers, fuse bases, and other similar parts, and shall be a minimum of class B (266 °F [130 °C]) in accordance with MIL-DTL-917. Insulating material shall be sealed to prevent the absorption of water. Primary conductor insulating material not in accordance with MIL-DTL-917 requirements shall be subjected to the flame testing requirements in accordance with MIL-DTL-917. Insulation temperature rating selection shall be confirmed based on temperature measurements in accordance with MIL-DTL-917 with an ambient temperature of 122 °F (50 °C).

3.2.9 Insulation resistance. Insulation resistance from conductor to conductor or conductor to ground shall be not less than 10 megohms at 77 °F (25 °C). User equipment shall tolerate the test voltage between each power conductor and ground without equipment damage, arc over, degradation, or abnormal operation and shall be subjected to temperature rise testing in accordance with MIL-DTL-917.

3.2.10 Transformers and inductors. Transformers and inductors shall meet the requirements of MIL-PRF-27. Alternate transformers and inductors may be used but shall meet the design, inspection, and test requirements for low quantity and non-QPL transformers and inductors in accordance with MIL-DTL-917.

3.2.11 Terminals and connections. The equipment shall be completely wired internally and to terminals provided for external connections, or with connectors in accordance with MIL-STD-2003. Terminals provided for external connections shall take cable connections up to 150 percent current rating of the equipment and shall be arranged on a terminal board located to provide ready accessibility for making connections. The terminals for external wiring shall be located in wiring compartments as specified in 3.2.6. Protective shields, as necessary, shall be furnished to protect exposed terminals or mounting boards within the equipment from damage due to penetration of the enclosure by moisture or foreign objects. Test points shall be provided for all control circuits for making measurements of voltage, current, and resistance as necessary for troubleshooting. All test points shall be available at terminals arranged on a terminal board located in a readily accessible position within the equipment enclosure. Terminal lugs shall be provided for all leads except for the main power input and power output leads. All connections using wire larger than 4 inches in diameter (4000 circular mils) shall be provided with efficient locking devices to prevent their becoming loosened by shock or vibration. Terminal boards shall be in accordance with MIL-DTL-917 except (a) where the terminal board is integral and considered a part of another approved part, and (b) where the application rating or space configuration makes their use impractical. Terminals shall be provided with contact surfaces of areas and materials that ensure the operation and freedom from oxidation of contact surfaces when the equipment is operated continuously in local temperatures of 230 °F (110 °C) as experienced by the terminal lugs. Threaded surfaces of bolts and studs shall not be utilized for current carrying contacts. Where wire smaller than 2 inches in diameter (2000 circular mils) is involved and the size of a part does not allow for the installation of screw type terminals, soldered connections are permitted. All such connections shall be in accordance with MIL-DTL-917.

3.2.12 Wiring. All wiring in the equipment shall be of sufficient capacity for rated duty of the equipment and shall be in accordance with the wire, wiring methods, marking, and bus bar requirements of MIL-DTL-917.

3.2.13 Dielectric strength. All equipment shall be designed and constructed to withstand a dielectric test voltage between separate circuits and ground for a period of 1 minute as specified in 4.5.17.

3.2.14 Ventilation. The equipment shall be cooled by natural convection without the use of fans. This shall be accomplished by the construction of enclosures and arrangement of parts to give a chimney-effect, natural draft. The dimensions of the equipment shall include any necessary allowance for air intake and discharge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

3.2.15 Weight and size. The maximum weight and size of each equipment shall be as specified in table I. Dimensions are overall, including screw heads and all protrusions, except for mounting straps (see 3.2.6).

TABLE I. Weight and size requirements for 440V equipment.

kVA	Maximum dimensions (inches)			Maximum weight (pounds)
	Height	Width	Depth	
15	33	20	16	250
30	28	30	16	400
22.5	32	28	13	350

3.2.16 Corrosion control. Equipment materials shall be selected or processed to provide corrosion resistance in accordance with the metals requirements of MIL-DTL-917. If powder-coated paint is used, it shall not affect the grounding of exposed equipment.

3.2.17 Creepage and clearance distances. Creepage and clearance distances shall be in accordance with the creepage and clearance distance requirements of MIL-DTL-917 with set C and enclosed spacing.

3.2.18 Heating. The equipment shall be able to meet the requirements of 3.3.2 when undergoing heating testing in accordance with MIL-DTL-917.

3.3 Design and construction.

3.3.1 Operation. Operation shall be accomplished by means of either static or moving parts. Each equipment of a specific rating and design shall be for use as a single-phase unit or as three identical units connected in a three-phase bank. Adjustments consisting of line drop compensation for resistance, inductance, and output voltage (see 6.4.9) shall be required to put the equipment into performance at the time of installation. The single-phase regulators shall operate in a three-phase bank when connected in a four-wire wye circuit with a grounded neutral or in a delta circuit. Output voltage compensation capability (see 3.3.4) shall be furnished in all equipment. Control of the equipment shall be accomplished by means of devices mounted within the basic equipment enclosure. The equipment shall be designed for continuous duty (see 6.4.1) under either single-phase or three-phase application. The equipment shall operate in accordance with 3.3.2 when wired into a three-phase configuration with forward or reversed phase rotation (see 6.4.11). Each equipment shall be designed with provisions for local sensing and remote sensing.

3.3.2 Operational configuration. The equipment shall be capable of regulating output voltage according to the following modes of operation:

a. Internal (or local) sensing and regulation of output voltage at the terminals of the equipment.

b. Remote sensing and regulation of voltage directly at the load or other remote point (output voltage compensation not used).

c. Internal (or local) sensing of output voltage at the terminals of the equipment with provision for output voltage compensation (see 3.3.4) for single-phase and three-phase wye operation.

3.3.2.1 Delta configuration. When connected in a three-phase 440V delta system, the equipment shall maintain output line-to-line voltages within the steady-state voltage regulation band specified in 3.3.3.1 (see 6.4.15 and 6.4.16).

3.3.2.2 Single-phase configuration. When operated as a single-phase unit in a 440V system, the equipment shall maintain output voltage within the steady-state voltage regulation band specified in 3.3.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

3.3.3 Voltage regulation (steady-state).

3.3.3.1 Sensing regulation. RMS output voltage shall remain regulated without adjustment for all conditions specified in table II. Steady-state output voltage shall be regulated within the following bands, corresponding to the mode of equipment operation:

a. ±0.5 percent of rated voltage (see 6.4.13) for operational modes specified in a and b of 3.3.2.

b. ±1.0 percent of rated voltage for the operational mode specified in 3.3.2.c.

TABLE II. Electrical operation characteristics.

Characteristic	Value
Input voltage	440V
Output voltage	200/115V
Rated voltages (RMS)	440V, single-phase 440V, line-to-line (three-phase delta)
Input voltage, steady-state band	±5 percent of rated voltage
Input/output frequency, steady-state	400 Hz ±½ percent
Output load variation	No load to rated full load, single-phase No load to rated full load, three-phase balanced
Output load power factor	0.5 lagging to 0.9 leading
Output load unbalance (maximum) (see 6.4.7)	
Three-phase wye (four-wire)	10 percent of rated output line current in one line Rated output line current in remaining two lines
Three-phase delta	20 percent of rated output line current in one line 80 percent of rated output line current in remaining two lines
Ambient temperature	73 to 122 °F

3.3.3.2 Three-phase delta configuration. Steady-state voltage regulation performance shall be as specified in table III for the operational modes of 3.3.2.a and b when the equipment is connected in a three-phase delta configuration under the conditions of unbalanced line current specified in table II.

TABLE III. Percent of rated input voltage for a three-phase delta configuration.

Phase AB	Phase BC	Phase CA
95	100	105
105	95	100
100	105	95

3.3.3.3 Three-phase wye configuration. Steady-state voltage regulation performance shall be as specified in table IV for the operational modes of 3.3.2.a and b when the equipment is interconnected in a three-phase, four-wire wye system (with grounded neutral) under the conditions of unbalanced line current specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

TABLE IV. Percent of rated input voltage for a three-phase wye configuration.

Phase AN	Phase BN	Phase CN
95	100	105
105	95	100
100	105	95

3.3.4 Output voltage compensation (three-phase, four-wire wye system). Each equipment shall be designed to provide means for output voltage compensation covering a minimum range extending from zero to +7.07 percent of rated output voltage at rated full load to compensate for voltage drop in the cabling between the equipment and the connected load. The compensation shall be provided in the form of separate resistive compensation and inductive compensation, each to be independently adjustable over a range from zero to 5 percent of rated voltage at rated full load. Any regulation error introduced by the specified output voltage compensation shall be constrained to permit the RMS output voltage at the end of the load cable to remain regulated within ±1 percent of rated voltage over a load range extending from no load to rated full load. The requirements on magnitude of output voltage compensation and permissible regulation error specified herein shall be satisfied when the equipment is operated in a three-phase, four-wire wye system (with grounded neutral) under conditions of balanced load. Compensation shall be linear as a function of load as required to meet the performance requirements of this paragraph. Compensation circuitry and parts shall be so designed that no combination of adjustments shall cause the regulator terminal output voltage to exceed 110 percent of rated voltage of any load.

3.3.5 Input/output voltage transient effect. If an undervoltage or overvoltage transient of maximum amplitude equal to 5 percent of rated peak voltage is superimposed upon the steady-state input voltage, the peak transient occurring in the output voltage shall not exceed 8 percent of rated peak output voltage. The output voltage shall recover to, and remain within, the steady-state regulation band specified in 3.3.3 within 0.1 second after the input voltage has recovered to the specified 5 percent steady-state band. The superimposed transient shall be induced from any input voltage value within the ±5 percent steady-state voltage band.

3.3.6 Output voltage transient. When the equipment, operated in accordance with 3.3.2.a, is subjected to the step removal of load specified in 4.5.7.2, the maximum overvoltage transient excursion occurring in the output voltage during the first six cycles after step change of load shall not exceed the peak magnitude of the input voltage transient by more than 10 percent of rated peak voltage. After six cycles following step load removal, the maximum overvoltage transient in output voltage shall not exceed the peak magnitude of the overvoltage transient in the input voltage by more than 5 percent of rated peak voltage. Whenever equipment operated in accordance with 3.3.2.a is subjected to the step applications of load specified in 4.5.7.2, the undervoltage transient excursion occurring in the output voltage after three cycles following step increase of load shall not exceed the peak magnitude of the input voltage transient by more than 5 percent of the rated peak voltage. Operation of the equipment shall be such that the input voltage prior to and following the load changes of 4.5.7.2 lies within the ±5 percent steady-state regulation band. The output voltage shall recover to, and remain within, the prescribed ±0.5 percent steady-state regulation band within 0.2 second after the input voltage has recovered to the specified ±5 percent steady-state band. No sustained oscillations shall develop.

3.3.7 Output voltage modulation. Calculated on a peak basis, any amplitude modulation (see 6.4.8) present in the output voltage of a regulator equipment shall not exceed 0.5 percent of rated peak voltage when the regulator is connected to a power source that exhibits less than 0.2 percent modulation, as determined by operating the source into a passive load having the same kVA loading effect as the regulator equipment.

3.3.7.1 Output voltage modulation gain. When the steady-state input voltage is amplitude modulated at an RMS amplitude equal to 2, 5, or 10 percent of rated RMS voltage and at a frequency ranging from 10 to 100 Hz superimposed upon the steady-state 400 Hz waveform, the maximum amplitude of the resulting output modulation in the steady-state output voltage waveform shall not exceed 150 percent of the corresponding input modulation when wye-connected regulator equipment is operated at rated input voltage, within the ±5 percent steady-state band, and at rated frequency. Modulation gain shall be confined to 180 percent when the regulator equipment is interconnected in three-phase delta banks and operated under the conditions specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

3.3.8 Induced voltage. Each equipment shall perform in accordance with the inspection requirements of MIL-DTL-917 during the application of an induced voltage.

3.3.9 Direct current (DC) resistance. Changes to DC resistance shall not exceed 3 percent when each equipment is tested at a room temperature of 68 °F (20 °C) versus an ambient temperature of 122 °F (50 °C) in accordance with the inspection requirements of MIL-DTL-917.

3.3.10 Polarity. Each equipment shall exhibit a polarity in accordance with the inspection requirements of MIL-DTL-917 during the application of a series voltage test.

3.3.11 No-load input excitation current characteristics. Each equipment shall exhibit a no-load input current excitation characteristic that shall remain within the constraints on amplitude and slope specified in 3.3.11.1 and 3.3.11.2.

3.3.11.1 No-load excitation current (steady-state). Under any steady-state operating condition, within an input voltage range extending from an undervoltage 12.5 percent below rated RMS voltage to an overvoltage point 10 percent above rated RMS voltage, the no-load, or excitation, regulator input line current, expressed as percent of rated full load output line current, shall be confined to the limits specified below:

a. Single-phase or three-phase wye (four-wire):
   (1) 3.5 percent for 150 ampere and 260 ampere regulators
   (2) 6 percent for 75 ampere regulators
   (3) 6 percent for 20 ampere and 40 ampere regulators
b. Three-phase delta:
   (1) 6 percent for 150 ampere and 260 ampere regulators
   (2) 10 percent for 75 ampere regulators
   (3) 10.5 percent for 20 ampere and 40 ampere regulators

3.3.11.1.1 Leading component of excitation current. Within the input voltage range of 3.3.11.1, the leading component of the input line current shall not exceed 1.5 percent of the maximum rated output line current for single-phase or three-phase, four-wire wye connections, and 3 percent of the maximum rated output line current for three-phase delta operation.

3.3.11.1.2 Initial energization limits. The limits on no-load input current specified in 3.3.11.1 shall apply after the first 10 cycles of input current following equipment energization.

3.3.11.2 No-load input current slope (steady-state). When the equipment is energized at no load, any tendency to exhibit a characteristic of increasing input current with decreasing input voltage shall not exceed 0.9 percent (of rated current) increase for each increment of 4.5 percent (of rated voltage) decrease. This characteristic shall be met for any increment in input voltage over the range extending from 12.5 percent below rated voltage to 10 percent above rated voltage under steady-state input conditions as specified in 3.3.11.1.

3.3.12 Output voltage build-up time. The output voltage of the equipment shall build up to a point corresponding to 85 percent of rated RMS voltage within 100 milliseconds after the energization of the equipment (see 6.4.10). After reaching the 85 percent voltage value, the output voltage shall continue to increase, approaching a point within a ±0.5 percent band of rated RMS voltage. Steady-state operation conforming to the voltage regulation requirements of this specification shall be attained within 250 milliseconds after energization. In no event shall the output voltage overshoot the rated voltage by more than 10 percent after 100 milliseconds following energization. Build-up of output voltage shall occur as specified under any load condition within the rating of the equipment.

3.3.13 Waveform distortion.

3.3.13.1 Harmonic distortion. When the equipment is operated in accordance with 4.5.8.1 under any condition of output load and load power factor specified in table II, the RMS summation of all harmonic (see 6.4.3) components present in the output voltage waveform produced by the equipment shall not exceed 3 percent of rated RMS voltage. No individual output harmonic shall exceed 2 percent of rated RMS voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

3.3.13.2 Deviation factor. The waveform deviation factor (see 6.4.2) shall not exceed 5 percent whenever the equipment is operated in accordance with table II and 3.3.2.a.

3.3.14 Efficiency.

3.3.14.1 kVA efficiency.

3.3.14.1.1 Wye configuration. The kVA efficiency (see 6.4.4) of each equipment connected in a three-phase, four-wire wye system shall be at least 70 percent at any load between 60 percent load and rated full load, at 0.75 lagging and unity power factors.

3.3.14.1.2 Delta configuration. The kVA efficiency of each equipment connected in a three-phase delta system shall be at least 60 percent at any load between 60 percent load and rated full load, at 0.75 lagging and unity power factors.

3.3.14.2 Kilowatt (kW) efficiency. The kW (power) efficiency (see 6.4.5) of each equipment shall be at least 90 percent at any load between 60 percent load and rated full load, at 0.75 lagging and unity power factors.

3.3.15 Endurance. The equipment shall withstand the test of 4.5.13.

3.3.16 Phase balance. When the equipment is connected in a three-phase delta configuration, it shall operate in accordance with steady-state condition numbers 28, 29, and 30 of table VII.

3.3.17 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Diagrams and identification plates.

3.4.1 Connection diagram. Each equipment shall contain a diagram of electrical connections with each terminal and lead designed to correspond to markings as shown on the equipment. In addition, a schematic circuit diagram and description of operation shall be included.

3.4.2 Identification and information labels. Identification labels shall be furnished and installed on all parts of the equipment and on the completed equipment. Information labels shall be furnished and installed on the complete equipment.

3.4.2.1 Labels for parts. Labels and the marking of part assemblies, such as rectifiers and similar items, shall comply with the applicable referenced specifications created by the manufacturer (see 6.7). Identification of other parts on the completed equipment, such as transformers and inductors, shall be made with the use of identification labels or by stamping or stenciling the required following minimum data:

a. Manufacturer’s name and identification number as shown on applicable drawings created by the manufacturer (see 6.7).

b. Title of part (that is, transformer, inductor, and similar items).

c. National Stock Number (NSN), if available.

If stamping or stenciling is done, it shall be applied and protected in such a manner as to be completely legible during the full life of the equipment. All labels shall be furnished as part of the equipment and shall be attached to parts of the equipment that will not ordinarily be renewed during its normal service life, in positions that are readily accessible and where they can be read at all times without danger to personnel.

3.4.2.2 Labels for the completed equipment. The identification label for the equipment shall indicate data as specified on figure 1. The information labels for the complete assembly shall indicate properly marked wiring diagrams showing connections for single-phase or three-phase use of the regulators in either a delta or wye system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

LINE VOLTAGE REGULATOR

TYPE: SPR_____ LINE AMPERES AC

PHASE: FREQUENCY _____ Hz

INPUT: _____ VOLTS AC

OUTPUT: _____ VOLTS AC

AMBIENT: 50 °C YEAR _____

CAT. NO.: _____ SERIAL NO.: _____

NSN: _____ CONTRACT: _____

DWG. NO.: _____ ACCEPTANCE STAMP

MANUFACTURER'S NAME _____

FIGURE 1. Line voltage regulator identification label.

3.5 Maintainability.

3.5.1 Primary corrective maintenance. The equipment shall be designed so that primary corrective maintenance (see 6.4.12) can be performed by the replacement of subassemblies. In the event that special tools are required for the removal or insertion of subassemblies, all such tools shall be supplied by the equipment manufacturer. Special tools are defined as those tools not listed in the GSA Global Supply Catalog. Equipment design should conform to the design-for-maintainability guidance of MIL-HDBK-338, MIL-HDBK-470, MIL-HDBK-472, and MIL-HDBK-2084 (see 6.6).

3.5.2 Secondary corrective maintenance. The subassemblies intended as replaceable items for primary corrective maintenance actions in case of equipment failure shall be designed to facilitate shipboard repair (see 6.4.14). Any subassembly repairs requiring higher echelon corrective maintenance should be so identified in technical manuals provided by the manufacturer (see 6.7). Except for irreplaceable parts, replaceable subassemblies shall be capable of repair by Navy personnel through the replacement of individual parts. Where special tools are required for subassembly repair, all such tools shall be furnished by the equipment manufacturer.

3.5.3 Spare and repair parts. Spare and repair parts should be furnished as specified (see 6.2). Spare or repair parts for the equipment covered by this specification should meet the same requirements as the parts used in the manufacture of the equipment.

3.6 Workmanship. All metal surfaces shall have a smooth finish and all details of manufacture, including the preparation of parts and accessories, shall be in accordance with the best practice for high quality electrical equipment. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of windings of transformers and inductors, marking of parts, plating, lacquering, riveting, clearance between connections, ruggedness, and suitability of enclosure.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

(1) Group A

(2) Group B

4.2 First article inspection. First article inspection shall be performed on line voltage regulators sampled in accordance with 4.2.1 and 4.2.2 when a first article sample is required (see 3.1). First article inspection shall be conducted at a laboratory acceptable to the Technical Authority (see 6.4.17) and shall consist of the examination and tests specified in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

TABLE V. First article and conformance inspections.

Test	First article inspection 1/	Conformance inspection		Reference paragraph	
		Group A 2/	Group B 1/	Requirement	Test
Vibration	X	--	X 3/	3.2.1	4.5.14
Shock	X	--	X 3/	3.2.2	4.5.15
Airborne noise	X	--	X	3.2.3	4.5.5
Electromagnetic interference	X	--	X 3/	3.2.4	4.5.23
System protection	X	--	X 3/	3.2.5	4.5.24
Mounting of equipment	X	--	X	3.2.6	4.5.25
Effectiveness of enclosure	X	--	X	3.2.7	4.5.1
Insulation	X	X	X	3.2.8	4.5.26
Insulation resistance	X	X	X	3.2.9	4.5.3
Transformers and inductors	X	--	X	3.2.10	4.5.16
General examination	X	X	X	3.2.11, 3.2.12, 3.2.16, 3.4, 3.6	4.4
Dielectric strength	X	--	X	3.2.13	4.5.17
Ventilation	X	--	X	3.2.14	4.5.27
Weight and size	X	--	X	3.2.15	4.5.18
Creepage and clearance	X	--	X	3.2.17	4.5.2
Heating	X	--	X	3.2.18	4.5.4
Operation	X	X	X	3.3.1	4.5.6.2
Operational configuration	X	X	X	3.3.2	4.5.6.2
Delta configuration	X	X	X	3.3.2.1	4.5.6.2.b(2)
Single-phase configuration	X	X	X	3.3.2.2	4.5.6.2.a(2)
Voltage regulation performance, steady-state	X	X	X	3.3.3, 3.3.3.1, 3.3.3.2	4.5.6
Three-phase wye configuration	X	X	X	3.3.3.3	4.5.6.2
Output voltage compensation	X	X	X	3.3.4	4.5.28
Input voltage transient	X	--	X	3.3.5	4.5.7.1
Output voltage transient	X	--	X	3.3.6	4.5.7.2
Output voltage modulation	X	--	X	3.3.7	4.5.6.4
Output voltage modulation gain	X	--	X	3.3.7.1	4.5.10
Induced voltage	X	--	X	3.3.8	4.5.19
DC resistance	X	X	X	3.3.9	4.5.20
Polarity	X	--	X	3.3.10	4.5.21
No-load input excitation current	X	--	X	3.3.11	4.5.11

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

TABLE V. First article and conformance inspections – Continued.

Test	First article inspection 1/	Conformance inspection		Reference paragraph	
		Group A 2/	Group B 1/	Requirement	Test
Slope of no-load input excitation current versus input voltage	X	--	X	3.3.11.1, 3.3.11.2	4.5.11.1
Output voltage build-up time	X	--	X	3.3.12	4.5.7.3
Waveform distortion	X	--	X	3.3.13	4.5.8
Efficiency	X	X	X	3.3.14	4.5.12
Endurance	X	--	--	3.3.15	4.5.13
Phase balance	X	--	X	3.3.16	4.5.9
Maintainability	X	--	--	3.5	4.5.22

NOTES:

1/ Group B and first article inspection testing shall be conducted under the phase conditions specified in table VI.

2/ Tests shall be conducted as single-phase equipment during group A testing.

3/ See 4.3.1.2.

TABLE VI. Phase conditions for group B and first article inspection tests. 1/

Test	Phase condition		
	Single-phase	Three-phase delta	Three-phase wye 1/
Vibration	X	--	--
Shock	X	--	--
Airborne noise	X	--	--
Electromagnetic interference	X	--	--
System protection	X	--	--
Mounting of enclosure	X	--	--
Effectiveness of enclosure	X	--	--
Insulation	X	--	--
Insulation resistance	X	--	--
Transformers and inductors	X	--	--
General examination	X	--	--
Dielectric strength	X	--	--
Ventilation	X	--	--
Weight and size	X	--	--
Creepage and clearance	X	--	--
Heating	X	--	--
Operation	X	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

TABLE VI. Phase conditions for group B and first article inspection tests [1/] – Continued.

Test	Phase condition		
	Single-phase	**Three-phase delta**	**Three-phase wye [1/]**
Operational configuration	X	X	X
Delta configuration	--	X	X
Single-phase configuration	X	--	--
Voltage regulation performance, steady-state	X	X	X
Three-phase wye configuration	--	X	X
Output voltage compensation	X	X	X
Input voltage transient	X	X	X
Output voltage transient	X	X	X
Output voltage modulation	X	X	X
Output voltage modulation gain	X	X	X
Induced voltage	X	X	X
DC resistance	X	X	X
Polarity	X	X	X
No-load input excitation current	X	X	X
Slope of no-load input excitation current versus input voltage	X	X	X
Output voltage build-up time	X	X	X
Waveform distortion	X	X	X
Efficiency	X	X	X
Phase balance	--	X	--

NOTE:

[1/] With the exception of voltage regulation performance (steady-state), three-phase wye operation is not required during group B tests.

4.2.1 Sampling for first article inspection of 440V equipment. For all ratings of the equipment for 440V operation, four identical equipment units corresponding to the following rating on a per phase basis shall be subjected to first article inspection in accordance with table V:

a. 440V

b. 400 Hz

c. 40 amperes

4.2.2 Sampling for first article inspection of single rating. For a single rating of the equipment, conforming to a particular voltage and current rating, four sample units representing the class of equipment shall be subjected to first article inspection in accordance with table V.

4.3 Conformance inspection. Conformance inspection shall consist of the group A and B examination and tests of table V.

4.3.1 Sampling.

4.3.1.1 Group A. Each production unit shall be subjected to the group A tests specified in table V to determine conformance with this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

4.3.1.2 Group B. Three sample units of each specific equipment rating shall be selected at random for the group B tests specified in table V on the first production contract or order for that specific equipment rating. Thereafter, every 36 months, the specific equipment rating shall be subjected to the group B tests specified in table V except for shock and vibration. A group B test shall be required after any change in design that affects the performance characteristics. If routine test data reveals variations beyond a normal manufacturing tolerance, the Technical Authority may require any or all of the group B tests be conducted on a particular equipment to demonstrate that it conforms to this specification. Group B test reports shall be as specified (see 6.7).

4.4 General examination. Each equipment shall be subjected to a thorough examination to ascertain that the material, workmanship, and design are in conformance with this specification. The fit of parts shall be observed with particular reference to the interchangeability of such parts as are likely to require replacement during the normal service life of the equipment. Specific observation shall be given to the following:

a. All wiring shall be checked for layout and proper identification.

b. All terminals shall be checked for proper marking.

c. Wire connections (soldered and unsoldered) shall be checked for security.

d. Arrangement and layout of parts shall be checked for conformance with applicable approved drawings from the manufacturer (see 6.7).

e. Miscellaneous hardware, mounting supports, and brackets shall be checked for proper corrosion-resistant material or treatment in accordance with the requirements for corrosion-resistant metals and the requirements for the treatment and processing of metals for corrosion resistance in MIL-DTL-917.

4.4.1 Spare and repair parts. All spare parts and repair parts shall be subjected to a careful examination to ascertain that the materials, workmanship, and finish conform fully to the manufacturer's drawings (see 6.7) as approved by the Technical Authority. The principal object of this examination shall be to determine if the repair parts are exact duplicates of those used in the equipment. If there is reason to doubt the ready interchangeability of the repair parts with the original equipment parts, a demonstration of such interchangeability shall be performed.

4.5 Test procedures.

4.5.1 Effectiveness of enclosure. The equipment enclosure shall be tested in accordance with the basic inspection requirements of MIL-DTL-2036 to determine conformance with 3.2.7.

4.5.2 Creepage and clearance. Creepage and clearance distances shall be demonstrated by actual measurement to be in accordance with the creepage and clearance distance requirements of MIL-DTL-917 with set C and enclosed spacing to determine conformance with 3.2.17.

4.5.3 Insulation resistance. Insulation resistance measurement shall be made to determine conformance with 3.2.9. The measurement of insulation resistance shall be made with all circuits of equal voltage above ground connected together. Circuits or groups of circuits of different voltage above ground shall be tested separately. The time of test voltage application shall be not less than 60 seconds. Measurements shall be made at a room ambient temperature of 73 °F (23 °C) with all circuits at that ambient temperature. The relative humidity and ambient temperature shall be recorded.

4.5.4 Heating.

4.5.4.1 Conditions. Heating tests shall be performed in accordance with the operational configuration of 3.3.2.a, with line voltage regulators interconnected in a balanced three-phase delta system at 105 percent rated input voltage under conditions of rated full load output line current (balanced) at unity power factor. Tests shall be conducted at 122 °F (50 °C) ambient temperature. Photographs, drawings, or descriptive text shall be used to illustrate the location of temperature measuring probes on the various power magnetic parts and semiconductor devices, as specified (see 6.7). Conformance with 3.2.18 shall be demonstrated.

4.5.4.2 Mounting of equipment. Heating tests shall be made only with the equipment assembled and mounted in the manner for which it is designed. Barriers shall be adjacent to the enclosure, 2 inches from the back and each side. Temperatures shall be taken on all coil windings (including power transformers and inductors) and semiconductors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

4.5.4.3 Temperature measurements. The method of temperature measurement shall be either the thermometer method or the embedded detector method in accordance with the temperature measurement requirements of MIL-DTL-917. When the thermometer method is used, the true temperature rise shall be determined by adding the value of the temperature gradient to the indicated temperature rise. The temperature gradient shall be predetermined prior to the assembly of the equipment by taking measurements of the maximum temperature of the parts under rated application load by both the resistance method and the thermometer method. The difference in measurements between the thermometer method and the resistance method shall be the value of the temperature gradient. When the embedded detector method is used, the true temperature rise shall be considered the indicated temperature rise of the detector element giving the maximum value. For power semiconductor devices and voltage reference devices, a thermocouple shall be affixed to the case or flange of the device with a small quantity of oil putty. The temperature measured shall not exceed the maximum case temperature permitted for the device under the conditions of voltage and current or power dissipation existing within the equipment. Maximum case temperature shall be obtained from appropriate curves or charts in the applicable MIL-PRF-19500 specification sheets or the device manufacturer's commercial specifications for each device; alternatively, the maximum case temperature may be calculated by subtracting the product of case-to-junction thermal impedance and power dissipation from the maximum allowed junction temperature. Thermal impedance values shall be obtained from charts in MIL-PRF-19500 or the device manufacturer's commercial specifications. Power dissipation may be calculated as the average product of instantaneous voltage and instantaneous current over one cycle of operation in periodic circuits or as the product of voltage and current in continuous circuits; alternatively, power dissipation may be obtained from appropriate curves in MIL-PRF-19500 or the device manufacturer's commercial specifications. The temperature measurements outlined above shall demonstrate that the power semiconductors and voltage reference diodes have been applied in accordance with MIL-DTL-917, and that device operation is consistent with assigned ratings in accordance with MIL-PRF-19500 specification sheets or commercial specifications on nonstandard devices approved in accordance with the ambient temperature range requirements of MIL-DTL-917.

4.5.4.4 Duration of test. The heating test shall be continued until the temperature rise of parts that can be observed during testing have attained a steady final value. The time duration shall be not less than 4 hours.

4.5.4.5 Acceptance criteria. The maximum permissible temperature (hot spot temperature) in accordance with the ambient temperature range requirements of MIL-DTL-917 for the class of insulation used and the maximum rated temperature assigned to the power semiconductors shall not be exceeded.

4.5.4.6 Ambient measurement. The ambient or room temperature shall be measured in accordance with the ambient temperature range requirements of MIL-DTL-917 by means of two or more thermometers placed at different points around and at the same level as the equipment and at a distance of 3 to 6 feet from the enclosure. The thermometer shall be inserted in oil-filled cups not less than 1 inch in internal diameter and 2 inches high. The thermometer shall be protected from drafts and from heat radiation from the equipment under test and other sources.

4.5.4.7 Corrections. In determining temperature rise, no corrections shall be made for barometric pressure or humidity.

4.5.5 Airborne noise. The equipment shall be examined for airborne noise performance in accordance with the testing procedures of MIL-STD-1474 to determine conformance with 3.2.3. The prescribed noise tests shall be completed on equipment operated without output voltage compensation at rated input voltage and frequency. The equipment shall be tested successively at no load and full load, at 0.75 lagging and unity power factor.

4.5.5.1 Group A. When steady-state voltage regulation tests are performed at test condition numbers 2 and 23 of table VII, in accordance with the group A procedure of 4.5.6.1, a broadband measurement of the sound pressure level shall be conducted by means of a microphone positioned at a single point 3 feet in front of the regulator equipment. Instrumentation used in the broadband measurement of the sound pressure level shall conform to the instrumentation requirements of MIL-STD-1474. The single point observation of broadband noise shall demonstrate that the equipment does not produce any acoustic noise emission in excess of 65 decibels. In the event that this noise level is exceeded, a noise test in accordance with 4.5.5 shall be conducted to determine conformance with the requirements of 3.2.3.

4.5.6 Voltage regulation. Equipment shall be tested as specified in 4.5.6.1 through 4.5.6.4 under all conditions of voltage, frequency, ambient temperature, and load within the specified rating range. The output voltage shall remain within the specified regulation limits without adjustment. Voltage regulation tests shall be conducted under the conditions specified in table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

TABLE VII. Conditions of line voltage, load, frequency, and temperature.

Condition number	Load	Power factor	Input voltage	Frequency	Ambient temperature (°F)
	(Balanced)		**(Balanced)**		
1	No load	--	Rated –5%	Rated	73
2	No load	--	Rated	Rated	73
3	No load	--	Rated +5%	Rated	73
4	10% load	Unity	Rated –5%	Rated	122
5	10% load	Unity	Rated	Rated	122
6	10% load	Unity	Rated +5%	Rated	122
7	20% load	0.75 lagging	Rated –5%	Rated –0.5%	122
8	20% load	0.75 lagging	Rated	Rated –0.5%	122
9	20% load	0.75 lagging	Rated +5%	Rated –0.5%	122
10	40% load	Unity	Rated –5%	Rated +0.5%	122
11	40% load	Unity	Rated	Rated +0.5%	122
12	40% load	Unity	Rated +5%	Rated +0.5%	122
13	60% load	0.75 lagging	Rated –5%	Rated	73
14	60% load	0.75 lagging	Rated	Rated	73
15	60% load	0.75 lagging	Rated +5%	Rated	73
16	80% load	Unity	Rated –5%	Rated –0.5%	73
17	80% load	Unity	Rated	Rated –0.5%	73
18	80% load	Unity	Rated +5%	Rated –0.5%	73
19	Full load	0.75 lagging	Rated –5%	Rated	122
20	Full load	0.75 lagging	Rated	Rated +0.5%	122
21	Full load	0.75 lagging	Rated +5%	Rated –0.5%	122
22	Full load	Unity	Rated –5%	Rated	73
23	Full load	Unity	Rated	Rated	73
24	Full load	Unity	Rated +5%	Rated	73
	(Unbalanced lines A, B, C)		**(Unbalanced line-to-neutral, AN, BN, CN)**		
25	Full load	Unity	Each line-to-neutral voltage within steady-state band of table II	Rated +0.5%	73
	Full load				
	10% load				
26	10% load	0.75 lagging		Rated	122
	Full load				
	Full load				
27	Full load	Unity		Rated –0.5%	73
	10% load				
	Full load				

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

TABLE VII. Conditions of line voltage, load, frequency, and temperature – Continued.

Condition number	Load	Power factor	Input voltage	Frequency	Ambient temperature (°F)
	(Unbalanced, lines A, B, C)		**(Unbalanced, line-to-line, AB, BC, CA)**		
28	80% load	0.75 lagging	Each line-to-line voltage within steady-state band of table II	Rated –0.5%	73
	80% load				
	20% load				
29	20% load	Unity		Rated +0.5%	73
	80% load				
	80% load				
30	80% load	Unity		Rated	122
	20% load				
	80% load				

4.5.6.1 Group A. The test for steady-state voltage regulation shall be performed in accordance with condition numbers 1, 2, 3, 22, 23, and 24 of table VII. Output voltage shall be measured at the output terminals of each equipment employing internal sensing without output voltage compensation.

4.5.6.2 Group B. The equipment shall be tested in accordance with the following procedure to determine conformance with 3.3.3.

a. The equipment shall be interconnected to form three-phase, four-wire wye systems and investigated under the following modes of regulation:

(1) Using the operational configuration specified in 3.3.2.a, output line-to-neutral voltages shall be measured at the regulator terminals under condition numbers 7 through 12, 16 through 18, 20, and 21 of table VII.

(2) Using the operational configuration specified in 3.3.2.c, output line-to-neutral voltages shall be measured at the load under condition numbers 4, 7, 10, 13, 16, 19, and 22 of table VII.

(3) Using the operational configuration specified in 3.3.2.a, output line-to-neutral voltages shall be measured at the regulator terminals under condition numbers 25, 26, and 27 of table VII.

b. The equipment shall be tested in a three-phase delta system under the following modes of regulation:

(1) Using the operational configuration specified in 3.3.2.b, output line voltages shall be measured at the load under condition numbers 1 through 6, 13 through 15, 19, and 22 through 24 of table VII.

(2) Using the operational configuration specified in 3.3.2.a, output line-to-line voltages shall be measured at the regulator equipment terminals under condition numbers 4, 7, 10, 13, 16, 19, and 22 of table VII.

4.5.6.3 Specification measurements. Readings shall be taken and recorded on input voltage and frequency, input current, power factor at regulator input terminals, ambient temperature, output voltage and output current for all of the specified conditions. Curves shall be plotted of output voltage versus output current for each specified voltage regulation test run of table VII.

4.5.6.4 Output voltage modulation. Peak and RMS values of input and output modulation voltages shall be recorded during the voltage regulation test specified in 4.5.6.2.b(1). A calibrated oscilloscope and camera attachment shall be employed to observe and record instantaneous input and output waveform modulation. Oscillograms of input and output voltage waveforms shall be prepared in such a manner as to scale and quality as to ensure readable and permanent traces. Voltage readings and oscillograms shall demonstrate that any amplitude modulation occurring in the output voltage is confined to the limits specified in 3.3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

4.5.7 Transient voltage regulation. The test for input voltage transient (see 6.4.18), output voltage transient, and output voltage build-up time performance shall be conducted with the equipment operated under the operational configuration of 3.3.2.a. Voltage measurements shall be performed at the output terminals of the equipment.

4.5.7.1 Input voltage transient. The equipment shall conform to the requirements of 3.3.5 when tested as follows:

a. A set of equipment wired in a three-phase, four-wire wye system shall be operated on a steady-state basis in accordance with condition numbers 3, 14, and 22 of table VII. An undervoltage transient of 5 percent (of rated peak voltage) shall be simultaneously induced in each line-neutral input voltage at any point of the cycle.

b. The set of equipment shall be operated in a three-phase delta configuration in accordance with the steady-state conditions specified by condition numbers 3, 14, and 22 of table VII. An overvoltage transient of 5 percent (of rated peak voltage) shall be simultaneously induced in each input line voltage on phases AB, BC, and CA at any point of the cycle.

4.5.7.2 Output voltage transient. The equipment shall conform to the requirements of 3.3.6 when tested as follows:

a. Input and output line-neutral voltage excursions shall be recorded for three-phase, four-wire wye systems in accordance with the following test sequence:

(1) The equipment shall be subjected to a balanced step increase in load from 10 percent load to rated full load accomplished in a single step, starting with the steady-state conditions specified for condition number 4 of table VII. This test shall be repeated for steady-state condition numbers 5 and 6 of table VII. The equipment shall be subjected to a balanced step decrease in load from rated full load to 10 percent load, starting with the steady-state conditions specified for condition number 19 of table VII. This test shall be repeated for steady-state condition numbers 20 and 21 of table VII.

(2) Equipment shall be operated on a steady-state basis in accordance with test condition numbers 25, 26, and 27 of table VII. The equipment shall be subjected to a step increase in load extending from 10 percent load to rated full load on the lightly loaded phase.

b. Repeat the tests specified in 4.5.7.2.a(1) except that the equipment shall be connected in three-phase delta for measurement of line-to-line voltage transients on phases AB, BC, and CA.

4.5.7.3 Output voltage build-up. The equipment shall be energized at condition numbers 1 and 22 of table VII. Waveform traces obtained on the basis of this test shall demonstrate conformance with 3.3.12.

4.5.7.4 Oscilloscope reading. Record voltage transients occurring at input and output terminals of the equipment and recovery time to steady-state conditions for input and output line-to-line voltages (for delta) and line-to-neutral voltages (for wye). Oscillograms of voltage waveforms shall demonstrate that the equipment conforms with 3.3.5, 3.3.6, and 3.3.12.

4.5.8 Waveform distortion.

4.5.8.1 Harmonic. The harmonic content (see 6.4.3.1) of the output voltage waveforms of the equipment shall be measured during the voltage regulation tests of 4.5.6.2.a(1) and 4.5.6.2.b(2) and the phase balance test of 4.5.9. The harmonic content of the output voltage waveform of the 400 Hz source used to provide power to the regulator equipment under test shall not exceed the limits specified below, when the source supplies a resistive load equal to the rating of the line voltage regulators:

a. 3 percent of rated RMS voltage, total harmonic distortion.

b. 2 percent of rated RMS voltage, for any single harmonic component.

Insertion of the regulator into the circuit involved in the tests of 4.5.6.2 and 4.5.9 shall not cause the total input harmonics to increase by more than 2 percent or cause any single harmonic to increase by more than 1 percent. The operating condition that produces the most significant harmonic distortion of the steady-state waveform shall be identified and recorded. When the conditions of maximum harmonic content have been determined, the percentage content of each RMS harmonic component shall be recorded up to the forty-first harmonic of the steady-state voltage waveform. The recorded test values shall not exceed the maximum percentages specified in 3.3.13 for single harmonic components and total harmonic content present in output voltage waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

4.5.8.2 Deviation factor. Equipment shall be operated in accordance with 3.3.2.a and tested at condition numbers 2, 14, and 23 of table VII. Input and output voltage waveforms shall be observed on an oscilloscope at each specified condition of operation. Scope observations of waveform shall demonstrate that the equipment conforms to 3.3.13.2. Source characteristics shall be consistent with 4.5.8.1.

4.5.9 Phase balance. While connected in three-phase delta and operating under steady state condition numbers 28, 29, and 30 of table VII, equipment operation shall conform to 3.3.2.a. The three-phase delta configuration shall be tested for voltage balance at equipment output terminals. Output line-to-line voltages shall be measured to determine conformance with 3.3.3.

4.5.10 Output voltage modulation gain. With rated RMS input voltage applied to the equipment in accordance with condition numbers 2, 5, 14, and 23 of table VII, the input voltage waveform shall be modulated, during separate tests, by a modulation voltage of amplitude equal to 2, 5, and 10 percent of rated voltage from a variable frequency source. Modulation frequency shall be gradually increased over the range 10 to 100 Hz. As the frequency of the modulation voltage is varied over the indicated range, any frequency at which the amplitude of the output modulation voltage reaches a peak shall be noted and recorded. If such peaking is observed as modulation frequency is raised from 10 to 100 Hz, the modulation gain of the equipment shall be measured at the frequency where the amplitude of modulation of the output voltage has reached a peak and shown a definite decrease with further increase of frequency. In the event that multiple peaks occur, the highest value of output modulation voltage shall apply. The amplitudes of the input and output voltage waveform modulations shall be recorded by means of calibrated modulation detection and waveform analysis instrumentation. A plot of RMS output voltage modulation gain versus modulation frequency shall be made for each specified test. Any output voltage modulation gain exhibited by the equipment shall be confined to the limits specified in 3.3.7.1.

4.5.11 No-load input current characteristics.

4.5.11.1 Steady-state. The equipment shall be operated at no load. RMS input voltage shall be slowly varied in maximum increments of 4.5 percent of rated voltage over a range extending from an undervoltage 12.5 percent below rated voltage to an overvoltage 10 percent above rated voltage. No-load RMS input current shall be plotted with respect to variation of RMS input voltage. No-load input current points shall be defined in increments of 0.5 percent of rated current along the vertical axis (ordinate) measured from input current monitored at rated voltage. Input voltage points shall be defined in increments of 4.5 percent of rated voltage along the horizonal axis (abscissa) in order to determine points on the excitation curve. The rated voltage point shall be marked on the horizonal axis of the graph of current versus voltage. The specified input voltage increments shall be measured from the rated voltage point. RMS input current and input voltage shall also be recorded in the form of a table. The no-load input current characteristic plotted during this test shall demonstrate conformance with 3.3.11.1 and 3.3.11.2.

4.5.11.2 Leading current component. The quadrature (90 degree) component of each no-load input current of 4.5.11.1 shall be measured during the excitation test and recorded in terms of RMS amperes. A plot of leading current versus input voltage shall be made (lagging current when present shall be shown as negative leading current). The maximum leading current shall not exceed the limits of 3.3.11.1.

4.5.12 Efficiency.

4.5.12.1 kVA efficiency. The kVA outputs and inputs of the equipment shall be measured at 60 percent load and 100 percent load, under conditions of unity and 0.75 lagging power factor loads with input voltage of 95, 100, and 105 percent of rated RMS value to determine conformance with 3.3.14.1.

4.5.12.2 kW efficiency. Input power furnished to the equipment and output power developed by the equipment shall be measured under the conditions of output loading and input voltage variation specified for testing of kVA efficiency to determine conformance with 3.3.14.2.

4.5.13 Endurance. Equipment connected in a delta three-phase bank supplying the unbalanced load specified in 4.5.9 at the conditions of unbalanced steady-state input line voltages specified in 4.5.10 shall be operated for a period of 720 hours without occurrence of part failure or operation that deviates from the requirements of this specification. After each 100 hours of operation, the initially impressed load shall be changed in a single step from the 20 percent – 80 percent – 80 percent configuration to a balanced rated line current condition. Unity power factor loads shall be impressed on the equipment. At each step change of load, the performance of the equipment shall be monitored to confirm operation in accordance with 3.3.3.2. The conditions of steady-state load imbalance specified in 4.5.9 shall then be restored. In the event of part failure during the prescribed test period, or evidence of

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

equipment operation that departs from the performance specified in 3.3.3, the endurance test shall be discontinued and the cause of part failure or equipment maloperation shall be determined. The failed part shall be replaced, and the entire 720-hour test program shall be repeated without regard to the duration of the test cycle preceding observation of part failure or equipment malfunction. The endurance test shall not be terminated until the equipment has demonstrated continuous operation for a period of 720 hours without failure of parts internal to the equipment. The causes of part failures observed during this test and any ensuing corrective action shall be supported by recommendations on equipment design changes and improved part selection necessary to prevent recurrence of the failures detected during endurance testing.

4.5.14 Vibration. Vibration tests shall be conducted in accordance with the type I requirements of MIL-STD-167-1. The test features specified for shock tests (see 4.5.15) shall apply also for vibration tests. The sample selected for vibration tests shall be the same sample selected for shock tests.

4.5.15 Shock. High-impact shock tests (electrical operation and mechanical damage) shall be conducted to determine conformance with 3.3.2. Test features shall be as follows:

a. General. The completely assembled equipment shall be checked for conformance with 3.3.2 and 3.3.3 following the shock test. Equipment rated at 260 amperes shall be energized and operated at 45 percent of rated load during the shock test. Equipment of all other ratings shall be energized and operated at rated full load during the shock test. Regulator output voltage shall be monitored to ensure continuity of operation under shock blow impact. Upon completion of the shock test, steady-state voltage regulation shall be examined in accordance with table VII to determine conformance with 3.3.3.

b. Definition of failure to perform principal functions.

(1) Breakage of any parts, including mounting bolts. Minor chipping of parts, such as plastic knobs and cases, and minor distortion of parts are permitted where such chipping or distortion cannot in any manner impair operation of the equipment as specified herein.

(2) Any distortion of parts, including enclosure and framework, which would prevent operation as specified herein.

(3) A value of insulation resistance lower than that specified (see 3.2.9).

(4) Low dielectric strength. After shock tests, the dielectric test shall be conducted at a voltage equal to 65 percent of the voltage specified in 4.5.17.1. Failure to pass this test shall be cause for rejection.

(5) Failure to pass visual examination. The equipment shall be carefully examined after removing all removable panels and doors to ascertain any mechanical damage. Partial disassembly shall be performed to aid in determining any possible damage when requested by the Technical Authority.

(6) Interruption of equipment operation by tripping of a circuit breaker or blowing of a fuse placed in the AC line, which supplies power to the regulator unit, as a direct effect of regulator maloperation caused by shock impact.

(7) Failure to perform any electrical tests requested by the Technical Authority during the following shock test. No adjustment or replacement of damaged parts shall be permitted during shock tests unless specifically approved by the Technical Authority.

c. Mounting. The equipment shall be mounted on the shock machines in a manner simulating shipboard installations in accordance with the mounting and testing requirements of MIL-DTL-901 for mounted equipment.

d. Disposal of shock-tested equipment. Disposal of shock-tested equipment shall be as specified in 6.5.

4.5.16 Transformers and inductors. Transformers and inductors that are not in accordance with MIL-PRF-27 shall be subjected to the Similarity Process of MIL-DTL-917 or tested in accordance with test numbers 1 through 14 of MIL-DTL-917 for low quantity and non-QPL-qualified transformers and inductors.

4.5.17 Dielectric strength. The dielectric test shall be in accordance with 4.5.17.1 through 4.5.17.3, shall precede shock and vibration testing, and shall be conducted again during the post shock testing of equipment (see 4.5.15.b(4)). If the insulation resistance of the windings is known to be lower than as specified in 3.2.9 due to dirt, moisture, or damage to windings, this shall be remedied before the application of the dielectric test voltage. The dielectric test shall be made upon the completely assembled equipment and not upon individual parts, except in the case of repair parts that require dielectric tests (see 4.4.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

4.5.17.1 Test voltage. The frequency of the testing voltage shall be not less than 60 Hz and shall approximate a true sine wave. The value of the test voltage shall be as specified in table VIII and shall be applied continuously for a period of 1 minute. All semiconductors shall be shorted out before applying the voltage.

TABLE VIII. Values of test voltages.

AC (RMS) input or output voltage	Test voltage (RMS)
60 and under	600
61 to 90	900
91 to 600	1000 plus twice rated
Over 600	2000 plus 2½ times rated

4.5.17.2 Measurement of test voltage. The measurement of the voltage used in dielectric tests shall be made by the voltmeter method whereby the meter derives its voltage from the high-voltage circuit either directly or by means of a voltmeter coil placed in the testing transformer, or through an auxiliary ratio transformer.

4.5.17.3 Points of application. The test voltage shall be successively applied between each electric circuit and all other electrical circuits and grounded metal parts not electrically connected to it.

4.5.17.4 Examination during and after test. During and after the dielectric test, transformers and inductors shall be examined for evidence of arcing, flashover, breakdown of insulation, and damage.

4.5.18 Weight and size.

4.5.18.1 Weight. The weight of the equipment and the weight of the onboard repair parts shall be taken and recorded separately. The weight of the equipment shall not exceed the value shown on applicable approved drawings from the manufacturer (see 6.7) or the maximum value specified in table I, whichever is least. The weight of the onboard repair parts shall be recorded on a “per set” basis. This weight shall not exceed the value shown on applicable approved drawings from the manufacturer by more than 2 percent.

4.5.18.2 Size. The overall dimensions of each equipment shall be taken and recorded. The dimensions shall not exceed the requirements specified in table I.

4.5.19 Induced voltage. Transformers and inductors shall be subjected to a voltage sufficient to cause twice the rated voltage to appear across any winding. The test voltage shall be applied to any winding for at least 4.5 seconds. Windings shall be grounded as they would be in service. The test frequency shall be selected by the manufacturer and shall be remote from any resonant frequency. Transformers and inductors shall be examined for evidence of continuous arcing, breakdown of insulation, and abrupt changes in the input current during induced voltage testing.

4.5.20 DC resistance. The DC resistance of windings in transformers and resistors shall be measured at, or corrected to, 68 °F (20 °C) and recorded.

4.5.21 Polarity. With transformer or inductor windings connected in series and with voltage applied to one of their windings, comparison shall be made between the sum of the voltages across individual windings and the voltage across the series of windings.

4.5.22 Maintainability. Equipment malfunction should be simulated in accordance with MIL-HDBK-470. Troubleshooting and repair procedures should be performed in accordance with any applicable technical manual supplied by the manufacturer (see 6.7). Equipment repair should be demonstrated in a manner consistent with MIL-HDBK-470.

4.5.23 Electromagnetic interference. Equipment shall be tested in accordance with MIL-STD-461, tests RE101, RE102, RS101, RS103, CS101, CE102, CS114, and CS116 to determine conformance with 3.2.4.

4.5.24 System protection. The equipment shall be tested in accordance with the requirements of the type III electric power system of MIL-STD-1399-300-1.

4.5.25 Mounting of equipment. The equipment mounting shall be tested in accordance with the basic inspection requirements of MIL-DTL-2036 to determine conformance with 3.2.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

4.5.26 Insulation. The equipment shall be tested in accordance with the insulation suitability requirements of MIL-DTL-917.

4.5.27 Ventilation. The equipment shall be tested in accordance with the external examination requirements of MIL-DTL-2036.

4.5.28 Output voltage compensation. The equipment shall be tested for condition numbers 5, 6, 20, 21, 28, and 29 of table VII. Pass/fail criteria for these tests shall fall within the type III voltage requirements for shipboard electric power systems in accordance with MIL-STD-1399-300-1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the general performance requirements for line voltage regulators. Line voltage regulators are used in shipboard electrical power distribution systems in order to provide regulated output power for sensitive loads. Due to the nature of shipboard use, naval line voltage regulators can be exposed to extreme weather and environmental conditions that commercial equipment is not designed for. Equipment manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained as approved by the Technical Authority. Line voltage regulators procured to this specification are form, fit, and function replacements for prior revisions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Input and output rated voltage.

c. When first article inspection is required (see 3.1).

d. Requirements for spare or repair parts (see 3.5.3).

e. Packaging requirements (see 5.1).

f. Requirements for shock-tested equipment (see 6.5).

g. Requirements for documentation (see 6.7).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Definitions. The following definitions apply to the various technical terms wherever such terms appear in this specification.

6.4.1 Continuous duty. Continuous duty is a requirement of service that the equipment operates at any constant load in the range of no-load to full load for an indefinitely long time.

6.4.2 Deviation factor. The deviation factor of the voltage wave is the ratio of the maximum difference between corresponding ordinates of the voltage wave and of the equivalent sine wave to 1.414 times the rated voltage when the waves are superimposed to make the maximum difference as small as possible. This does not apply to ordinates within 10 electrical degrees on either side of zero voltage.

$$\text{Deviation factor } \% = \frac{\text{Max. deviation } (100)}{\sqrt{2}\ \text{nominal voltage}}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

6.4.3 Harmonic. A harmonic is a multiple of any particular frequency. Specifically, the "second harmonic" of a fundamental frequency is equal to twice that fundamental frequency.

6.4.3.1 Harmonic content. The harmonic content of a voltage wave is the ratio, expressed as a percent of the effective value, of the residue after the elimination of the fundamental, to the effective value of the equipment RMS voltage rating.

6.4.4 Kilovolt ampere (kVA) efficiency. The term kVA efficiency is defined as the ratio of kVA output to kVA input.

6.4.5 Kilowatt (kW) efficiency. The term kW efficiency represents the ratio of kW output, measured at any output load and load power factor specified in table II, to kW input.

6.4.6 Line voltage regulator. A line voltage regulator is a completely static device with regulation accomplished by means that may include the arrangement of transformers, inductors, resistors, capacitors, silicon rectifiers, silicon-controlled rectifiers, and silicon transistors (see 6.4.19). Reference to "equipment" is considered as applying to line voltage regulators covered in this specification.

6.4.7 Load unbalance. Load unbalance is the difference between the kVA load on any two phases expressed in percent of the single-phase rating.

6.4.8 Modulation amplitude. Modulation amplitude is a periodic sine wave voltage variation about rated equipment voltage expressed as follows:

$$\text{Modulation } \% = \frac{\text{E max} - \text{E min}}{\text{E max} + \text{E min}} \times (\text{peak or RMS})$$

6.4.9 Output voltage. Output voltage refers to the voltage at the point of regulation, which under modes b and c of 3.3.2, is the voltage at the load terminals as opposed to the regulator terminal voltage.

6.4.10 Output voltage build-up time. Output voltage build-up time is the time required for the load voltage to reach and remain within a specified band about rated load voltage, after the application of input power by a switch and the return of the input voltage to the steady-state input voltage band.

6.4.11 Phase rotation. The forward phase rotation, when facing a drawing, is AB, BC, and CA from top to bottom and AB, BC, and CA from left to right.

6.4.12 Primary corrective maintenance. Primary corrective maintenance is that maintenance performed on an as-required basis to return an equipment to service in the shortest possible time after a failure or malfunction has occurred.

6.4.13 Rated voltage. The term "rated voltage" appearing in this specification refers to identification plate RMS voltage, expressed as input volts (AC) and output volts (AC), such as 440 volts RMS.

6.4.14 Secondary corrective maintenance. Secondary corrective maintenance is a maintenance action, performed independently of primary corrective maintenance, which restores a malfunctioning subassembly removed from the equipment to the status of an operating spare.

6.4.15 Steady-state frequency. The steady-state frequency limits cover the maximum frequency variations expressed in percent of rated system frequency steady-state load conditions. This includes variations caused by load changes and environment, such as temperature, inclination, and meter error, but does not include transient changes.

6.4.16 Steady-state voltage regulation band. The steady-state voltage regulation band is the maximum RMS voltage variation expressed in percent of rated RMS equipment voltage under steady-state load conditions. This includes variations caused by load changes, voltage changes, frequency changes, and environment, such as temperature, inclination, and meter error, but does not include transient load changes or amplitude modulation effects.

6.4.17 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

6.4.18 Transient voltage. A transient voltage is the rapidly changing condition of the voltage which goes beyond the steady-state voltage limits and returns to the steady-state voltage band within a specified time period (recovery time). Maximum permissible transient voltage variation is expressed in percent of rated voltage.

6.4.19 Transistor. A transistor is an active semiconductor device with three or more electrodes.

6.5 Disposal of shock-tested equipment. When specified (see 6.2), equipment that has undergone shock testing should be accepted for use or disposed of as follows:

a. Equipment that has been subjected to the high-impact shock test under qualification inspection should be returned to the manufacturer for disposal, or upon request of the manufacturer, for disposal by the Government.

b. Equipment that has been subjected to the high-impact shock test under group B testing should be accepted as a production unit of the contract or order only under the following conditions:

(1) That any damaged parts are replaced.

(2) That post shock electrical tests show conformance with MIL-DTL-901. The extent of electrical testing after shock tests for specified operation should be as determined by the Technical Authority (see 4.5.15.b(7)) in each instance and should be dependent on the nature of required performance and inspection results of shock tests.

(3) That an identification plate is added identifying the unit as “high-impact shock tested”.

(4) That the unit is subjected to the same guarantee as other production units.

(5) That not more than one shock-tested equipment per contract or order is being offered for acceptance following the high-impact shock test.

6.6 Management control system documents. The following management control system documents should be included on DD Form 1660:

a. MIL-HDBK-470 (see 3.5)

b. A-A-59877

6.7 Recommended documentation. The following documentation should be requested as specified (see 6.2):

a. Manufacturer’s drawings

b. Maintenance drawings

c. Technical manuals

d. Test and examination reports

e. Oscillograms of voltage waveforms (see 4.5.7.4)

f. Certification drawings (see figure 2)

6.8 Subject term (key word) listing.

Capacitor
Clearance
Corrosion
Creepage
Electromagnetic interference (EMI)
Harmonic distortion
Inductive compensation
Linear
Local sensing
Natural convection
Output voltage modulation
Peak output voltage
Phase rotation
Regulation
Remote sensing

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

Resistive compensation
Semiconductor
Steady-state operation
Three-phase delta configuration
Three-phase wye configuration
Transformer
Voltage compensation
Voltage transient
Waveform distortion

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

A. Supplementing NAVSEA drawing:

Equipment	NAVSEA Dwg. No.	Rev.	Mfr's Drawing No.

B. Applicable specification _____ of _____ and Amend. of _____

C. Navy contract or shipbuilders order no. _____ Item _____ Date _____

D. Mfr. _____ Mfr's Order No. _____

E. Mfr's identification and rating:

Descriptive title _____ Type: _____ Freq: _____

Input: Voltage _____ Output: Voltage _____ Current _____

F. Application _____

G. No. of ships involved

H. Ships identification numbers.

I. No. of units: Per Ship _____, Per contract _____, For stock _____

J. No. of sets onboard repair parts; Per ship _____, Per contract _____

For stock _____

Note: See sheet 2 for listing of one set of onboard repair parts.

K. Wt. of onboard repair parts (unboxed): Per set _____, Per contract _____

L. Reference drawings

Title	NAVSEA Dwg. No.	Mfr's Drawing No.	Certification data

M. Notes:

N.

(Other listings as desired for Mfrs. information)

O.

<table>
<tr><td>Date:</td><td>CERTIFICATION DATA FOR</td><td colspan="2">MFR.</td></tr>
<tr><td>Drawn by:</td><td rowspan="5">(Equipment title)</td><td colspan="2">Mfr. Dwg. No.</td></tr>
<tr><td>Approved Date</td><td>NAVY DEPT.
NAVSEA DWG. NO.</td><td rowspan="4">Rev.</td></tr>
<tr><td>By</td></tr>
<tr><td>By</td></tr>
<tr><td rowspan="2">Approval reference ltr.</td></tr>
<tr><td></td><td>Wt. Calc. _____ Actual _____</td><td colspan="2">Cont'd. on Sht. 2, Sht. No. 1</td></tr>
<tr><td></td></tr>
</table>

FIGURE 2. Certification drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

List of onboard repair parts and tools						
Item No.	No. per set	No. per equipment	Name of part or tool	Identification data		Federal stock No.
				Contractor	Manufacturer	

Date:	CERTIFICATION DATA FOR	MFR.	
Drawn by:	(Equipment title)	Mfr. Dwg. No.	
Approved Date		NAVY DEPT. NAVSEA DWG. NO.	Rev.
By			
By			
Approval reference ltr.	Wt. Calc. _____ Actual _____	Cont'd. on Sht. ___, Sht. No. 2	

FIGURE 2. Certification drawing – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – GS

Preparing activity:
Navy – SH
(Project 6110-2024-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:33Z
Check the source to verify that this is the current version before use.